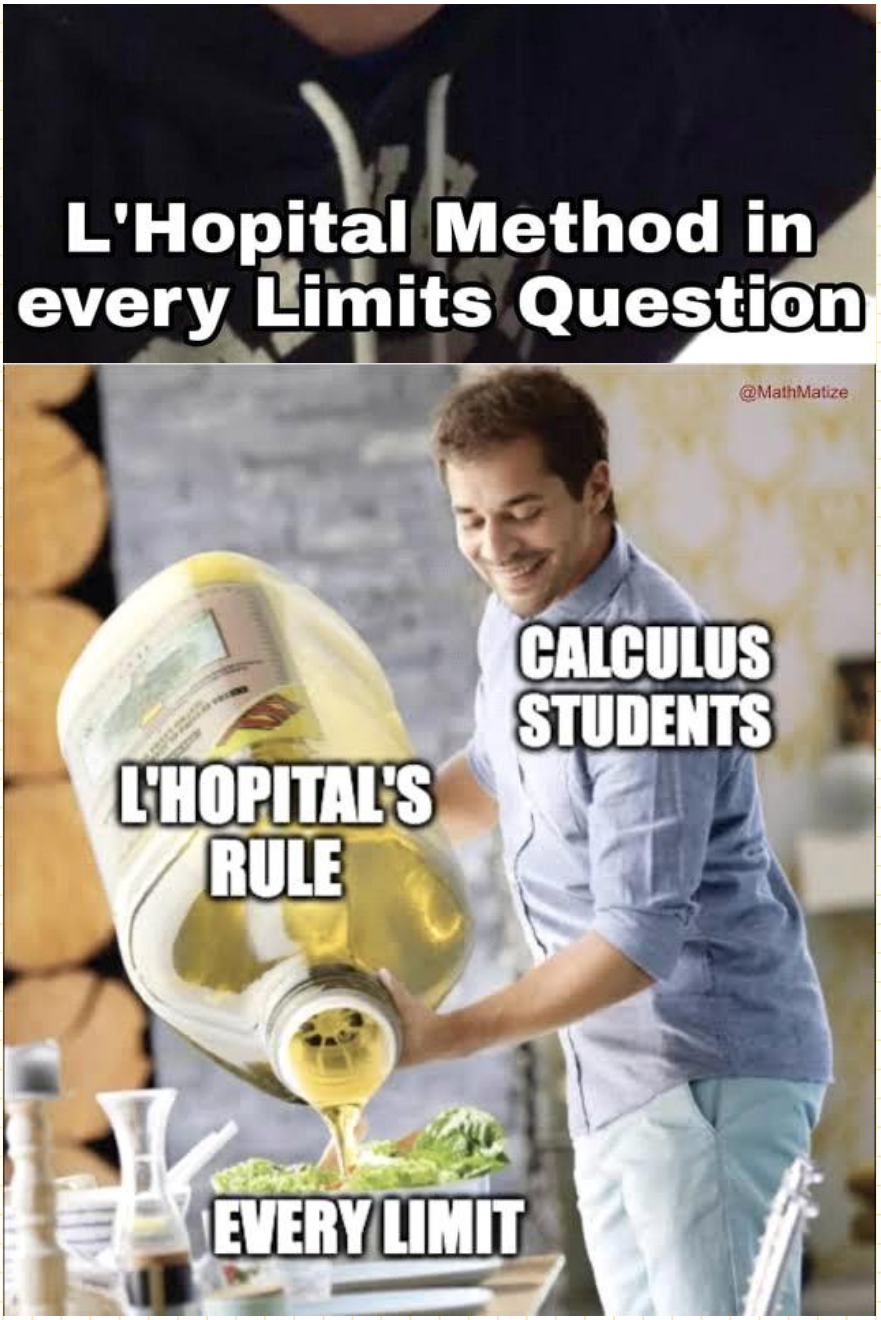
Application of Derivatives

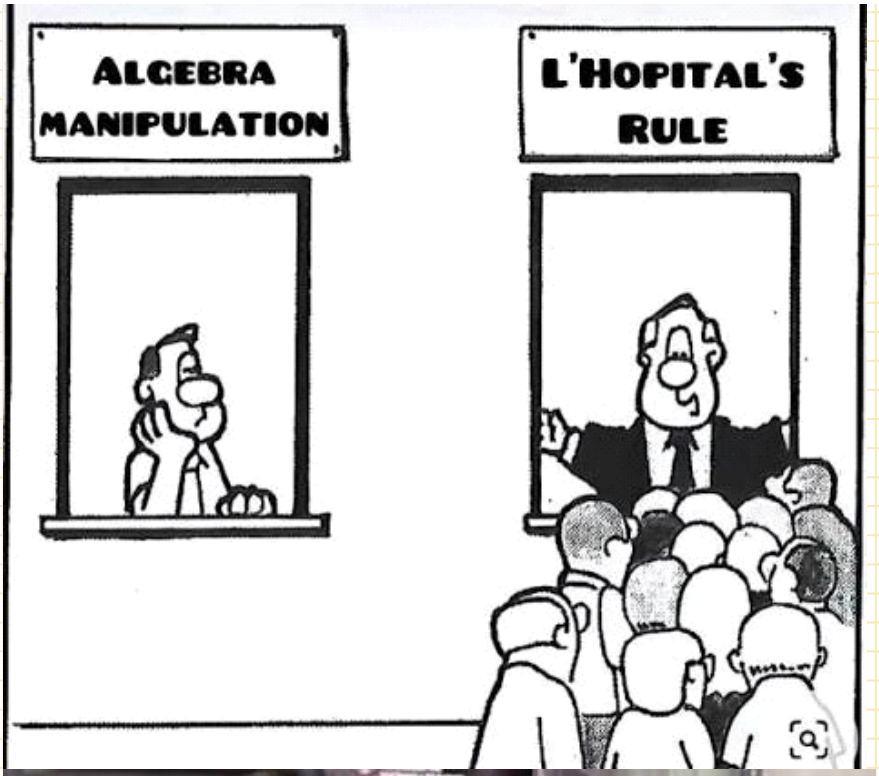
Monday, 19 October 2026 4:01 pm








L'Hopital Method in every Limits Question
@MathMatize
CALCULUS STUDENTS
L'HOPITAL'S RULE
EVERY LIMIT







Determinate-Indeterminate Forms Table

Indeterminate Forms	Determinate Forms
$0/0$	$\infty + \infty = \infty$
$+\infty/+\infty$	$-\infty - \infty = -\infty$

$0/0$	$\infty + \infty = \infty$
$\pm\infty/\pm\infty$	$-\infty - \infty = -\infty$
$\infty - \infty$	$0^{\infty} = 0$
$0(\infty)$	$0^{-\infty} = \infty$
0^0	$(\infty) \cdot (\infty) = \infty$
1^{∞}	
∞^0	
Use L'Hôpital's Rule	Do *Not* Use L'Hôpital's Rule

HISTORICAL BIOGRAPHY

Guillaume François Antoine de l'Hôpital
(1661–1704)

Johann Bernoulli
(1667–1748)

Indeterminate Form 0/0

If we want to know how the function

$$F(x) = \frac{x - \sin x}{x^3}$$

THEOREM 6—L'Hôpital's Rule Suppose that $f(a) = g(a) = 0$, that f and g are differentiable on an open interval I containing a, and that $g'(x) \neq 0$ on I if $x \neq a$. Then

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)},$$

assuming that the limit on the right side of this equation exists.

(a) $\lim_{x \to 0} \frac{3x - \sin x}{x} =$

b) $\lim_{x \to 0} \frac{\sqrt{1 + x} - 1}{x} =$

$\sqrt{1 + x} - 1 - x/2$

(c) $\lim_{x\to 0} \dfrac{\sqrt{1+x} - 1 - x/2}{x^2}$

$\lim_{x\to 0} \dfrac{1 - \cos x}{x + x^2}$

(a) $\lim_{x\to 0^+} \dfrac{\sin x}{x^2}$

$= \lim_{x\to 0^+} \dfrac{\cos x}{2x} = \infty$

(b) $\lim_{x\to 0^-} \dfrac{\sin x}{x^2}$

$= \lim_{x\to 0^-} \dfrac{\cos x}{2x} = -\infty$

EXAMPLE 4 Find the limits of these ∞/∞ forms:

(a) $\lim_{x\to \pi/2} \dfrac{\sec x}{1 + \tan x}$ **(b)** $\lim_{x\to\infty} \dfrac{\ln x}{2\sqrt{x}}$ **(c)** $\lim_{x\to\infty} \dfrac{e^x}{x^2}$.

EXAMPLE 5 Find the limits of these $\infty \cdot 0$ forms:

(a) $\lim_{x\to\infty} \left(x \sin \dfrac{1}{x}\right)$ **(b)** $\lim_{x\to 0^+} \sqrt{x} \ln x$

Solution

a. $\lim_{x\to\infty} \left(x \sin \dfrac{1}{x}\right) = \lim_{h\to 0^+} \left(\dfrac{1}{h} \sin h\right) = \lim_{h\to 0^+} \dfrac{\sin h}{h} = 1$

b. $\lim_{x\to 0^+} \sqrt{x} \ln x = \lim_{x\to 0^+} \dfrac{\ln x}{1/\sqrt{x}}$

$= \lim_{x\to 0^+} \dfrac{1/x}{-1/2x^{3/2}}$

$= \lim_{x\to 0^+} \left(-2\sqrt{x}\right) = 0$

EXAMPLE 6 Find the limit of this $\infty - \infty$ form:

$$\lim_{x \to 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right).$$

$$\frac{1}{\sin x} - \frac{1}{x} = \frac{x - \sin x}{x \sin x}.$$

$$\begin{aligned} \lim_{x \to 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right) &= \lim_{x \to 0} \frac{x - \sin x}{x \sin x} \\ &= \lim_{x \to 0} \frac{1 - \cos x}{\sin x + x \cos x} \\ &= \lim_{x \to 0} \frac{\sin x}{2 \cos x - x \sin x} = \frac{0}{2} = 0. \end{aligned}$$

If $\lim_{x \to a} \ln f(x) = L$, then

$$\lim_{x \to a} f(x) = \lim_{x \to a} e^{\ln f(x)} = e^L.$$

Here a may be either finite or infinite.

EXAMPLE 7 Apply l'Hôpital's Rule to show that $\lim_{x \to 0^+} (1 + x)^{1/x} = e$.

$$\ln f(x) = \ln (1 + x)^{1/x} = \frac{1}{x} \ln (1 + x),$$

$$\begin{aligned} \lim_{x \to 0^+} \ln f(x) &= \lim_{x \to 0^+} \frac{\ln (1 + x)}{x} \\ &= \lim_{x \to 0^+} \frac{\frac{1}{1 + x}}{1} \\ &= \frac{1}{1} = 1. \end{aligned}$$

Therefore, $\lim_{x \to 0^+} (1 + x)^{1/x} = \lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} e^{\ln f(x)} = e^1 = e.$

EXAMPLE 8 Find $\lim_{x \to \infty} x^{1/x}$.

$$\ln f(x) = \ln x^{1/x} = \frac{\ln x}{x},$$

l'Hôpital's Rule gives

$$\lim_{x\to\infty} \ln f(x) = \lim_{x\to\infty} \frac{\ln x}{x} \qquad \frac{\infty}{\infty}$$

$$= \lim_{x\to\infty} \frac{1/x}{1} \qquad \text{l'Hôpital's Rule applied}$$

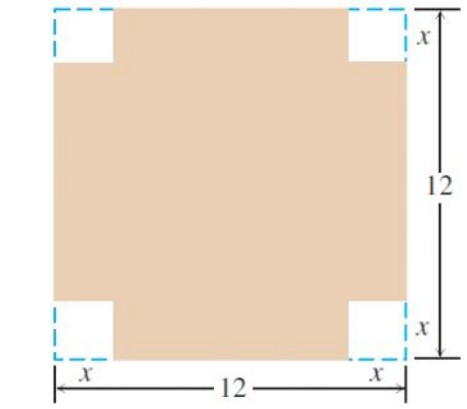
$$= \frac{0}{1} = 0.$$

Therefore $\lim_{x\to\infty} x^{1/x} = \lim_{x\to\infty} f(x) = \lim_{x\to\infty} e^{\ln f(x)} = e^0 = 1.$

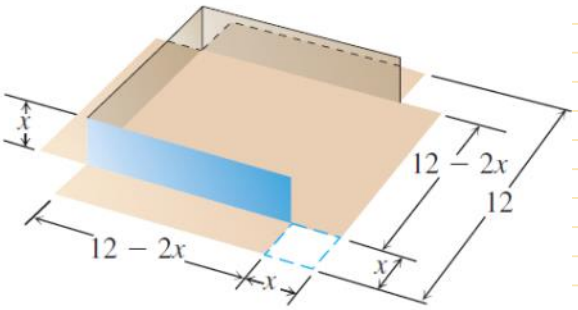
Applied Optimization

Solving Applied Optimization Problems

1. *Read the problem.* Read the problem until you understand it. What is given? What is the unknown quantity to be optimized?
2. *Draw a picture.* Label any part that may be important to the problem.
3. *Introduce variables.* List every relation in the picture and in the problem as an equation or algebraic expression, and identify the unknown variable.
4. *Write an equation for the unknown quantity.* If you can, express the unknown as a function of a single variable or in two equations in two unknowns. This may require considerable manipulation.
5. *Test the critical points and endpoints in the domain of the unknown.* Use what you know about the shape of the function's graph. Use the first and second derivatives to identify and classify the function's critical points.



(a)



EXAMPLE 1 An open-top box is to be made by cutting small congruent squares from the corners of a 12-in.-by-12-in. sheet of tin and bending up the sides. How large should the squares cut from the corners be to make the box hold as much as possible?

$$V(x) = x(12 - 2x)^2 = 144x - 48x^2 + 4x^3. \qquad V = hlw$$

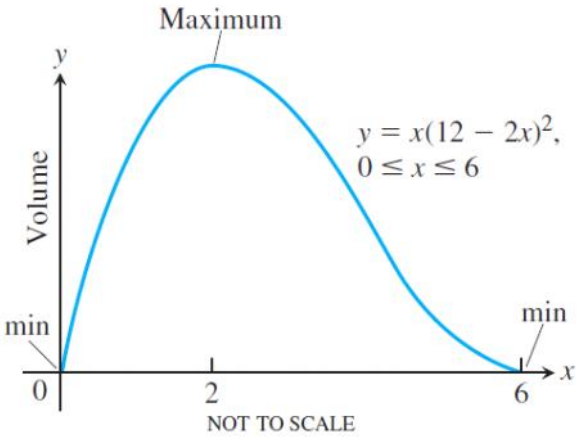


FIGURE 4.37 The volume of the box in Figure 4.36 graphed as a function of x.

$$\frac{dV}{dx} = 144 - 96x + 12x^2 = 12(12 - 8x + x^2) = 12(2 - x)(6 - x).$$

Critical point value: $V(2) = 128$

Endpoint values: $V(0) = 0, \quad V(6) = 0.$

EXAMPLE 2 You have been asked to design a one-liter can shaped like a right circular cylinder (Figure 4.38). What dimensions will use the least material?

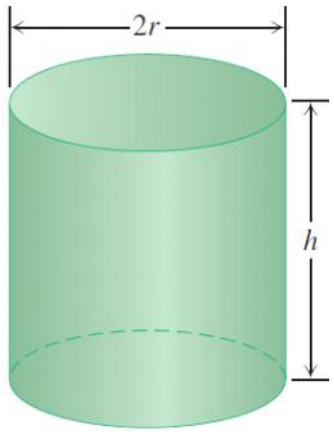


FIGURE 4.38 This one-liter can uses the least material when $h = 2r$ (Example 2).

$$\pi r^2 h = 1000. \qquad 1 \text{ liter} = 1000 \text{ cm}^3$$

Surface area of can: $A = \underbrace{2\pi r^2}_{\text{circular ends}} + \underbrace{2\pi rh}_{\text{cylindrical wall}}$

$$h = \frac{1000}{\pi r^2}.$$

$$\begin{aligned} A &= 2\pi r^2 + 2\pi rh \\ &= 2\pi r^2 + 2\pi r\left(\frac{1000}{\pi r^2}\right) \\ &= 2\pi r^2 + \frac{2000}{r}. \end{aligned}$$

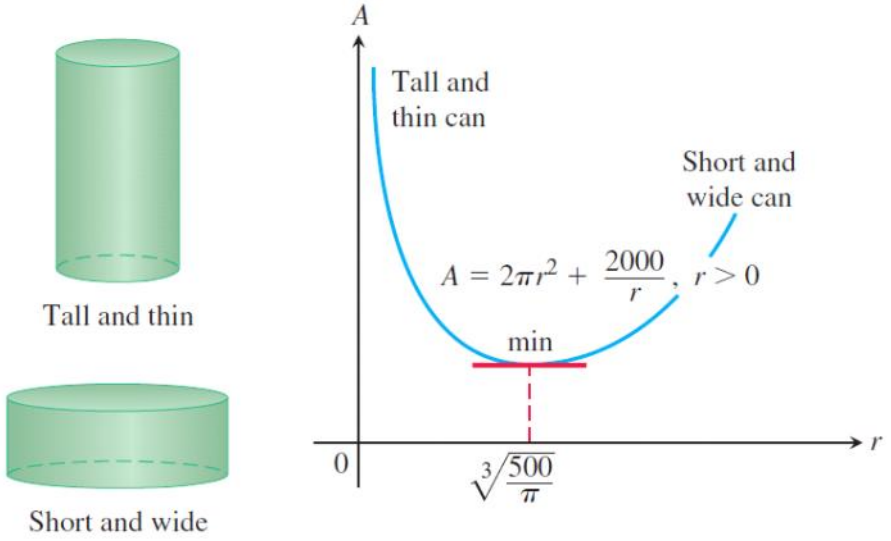


FIGURE 4.39 The graph of $A = 2\pi r^2 + 2000/r$ is concave up.

$$\frac{dA}{dr} = 4\pi r - \frac{2000}{r^2}$$

$$0 = 4\pi r - \frac{2000}{r^2}$$

$$4\pi r^3 = 2000$$

$$r = \sqrt[3]{\frac{500}{\pi}} \approx 5.42$$

What happens at $r = \sqrt[3]{500/\pi}$?

The second derivative

$$\frac{d^2A}{dr^2} = 4\pi + \frac{4000}{r^3}$$

$$h = \frac{1000}{\pi r^2} = 2\sqrt[3]{\frac{500}{\pi}} = 2r.$$

Problem statement

A chemical engineering company produces a chemical product through a reaction in a batch reactor. The production rate depends on the operating temperature T, measured in °C.

Based on experimental data, the company models the net profit per batch by

$$P(T) = -2T^2 + 240T - 5000,$$

where $P(T)$ is the net profit in Pakistani rupees (PKR) per batch, and the operating temperature is restricted to

$$40 \leq T \leq 80.$$

The quadratic model accounts for the combined effects of production efficiency and operating costs over the specified temperature range.

The objective is to determine the operating temperature that maximizes the net profit per batch.

Questions for students

1. Calculate the first derivative $P'(T)$ and explain its physical meaning in this context.
2. Determine the critical point by solving $P'(T) = 0$.
3. Use the second derivative test to determine whether the critical point gives a maximum or minimum profit.
4. Calculate the net profit at the optimal temperature.
5. Calculate the profit at the boundary temperatures $T = 40^\circ\text{C}$ and $T = 80^\circ\text{C}$. Verify that the critical point gives the greatest profit on the specified interval.
6. Explain why operating at a higher temperature does not necessarily lead to a higher profit in a chemical manufacturing process.

Applied Optimization: CSTR Operating Conditions

A chemical plant produces B through the first-order reaction $A \rightarrow B$ in a continuous stirred-tank reactor (CSTR). The reaction rate constant depends on temperature T (°C):

$$k(T) = 0.02e^{0.04(T-60)} \text{ min}^{-1}.$$

The conversion and operating cost are given by

$$X(T, \tau) = \frac{k(T)\tau}{1 + k(T)\tau},$$

$$C(T, \tau) = 10(T - 40)^2 + 50\tau,$$

where τ is the residence time in minutes and C is the operating cost in illustrative cost units.

The company requires a conversion of exactly 80%, with $40 \leq T \leq 80$.

Tasks

1. Formulate the two-variable constrained optimization problem to minimize $C(T, \tau)$.
2. Using $X = 0.8$, eliminate τ and express it as a function of T.
3. Obtain a single-variable cost function $C(T)$.
4. Find the critical point using the first derivative.
5. Apply the second derivative test and verify the global minimum on the feasible interval.
6. Determine the optimal temperature, residence time, and minimum operating cost.